

Module – 1: Introduction

(Introduction to Information Theory, Information rate and entropy, Measure of Information, Properties of entropy of a binary memory less source.)

- The performance of the communication system is measured in terms of its error probability.
- An error-less transmission is possible when the probability of error at the receiver approaches zero.
- Information theory is used for mathematical modeling and analysis of the communication systems.

1.1 What is information theory?

- *Information theory* is needed to enable the communication system to carry information (signals) from sender to receiver over a communication channel.
 - it deals with mathematical modelling and analysis of a communication system.
 - its major task is to answer to the questions of *signal compression* and *transfer rate*.
- Those answers can be found and solved by entropy and channel capacity.
- *Information Theory* is the fundamental theory behind *information manipulation*, including *data compression* and *data transmission*.

Q1: Comments on the information content of the following message:

Tomorrow the sun will rise from the east.

The phone will ring in the next one hour.

It will snow fall in Delhi this winter.

Ans.

- The first statement does not carry any information since it is sure that sun always rises from East. The probability of occurrence of first event is high or sure. Hence it carries less or negligible information.
- In the third statement predicts about phone ring in the time span of one hour. It does not mention exact time but span of one hour is mentioned. Hence it carries moderate information.
- In the winter season snow fall in Delhi is very rare. Hence probability of occurrence of this event is very rare, so it carries large amount of information.

1.2 Information

Information in information theory refers to the measure of uncertainty that occurs when an event or symbol is observed. A source is said to be discrete if it emits symbols from a finite or countable set. A source is memoryless if each symbol generated is statistically independent of all previously generated symbols. The source that satisfies both these condition is called as a discrete memoryless source (DMS). Consider such a DMS emitting discrete symbols from the set of fixed alphabets, i.e., $X = \{x_1, x_2, \dots, x_N\}$, with probabilities:

$$P(X = x_i) = p_i, i = 1, 2, \dots, N$$

This set of probabilities satisfies the following condition:

$$\sum_{i=1}^N p_i = 1$$

The amount of information transmitted through the symbol x_i is given as:

$$I_i = \log_2 \left(\frac{1}{p_i} \right)$$

1.3 Units of Information:

$$I_2(x_i) = \log_2 \left(\frac{1}{p_i} \right) \text{ bits}$$

$$I_e(x_i) = \log_e \left(\frac{1}{p_i} \right) \text{ nats}$$

$$I_{10}(x_i) = \log_{10} \left(\frac{1}{p_i} \right) \text{ Hartleys}$$

1.4 Entropy

Entropy is a fundamental concept in information theory that measures the average amount of information, uncertainty, or surprise associated with the outcome of a random variable. Entropy is defined as the average information content of a source, which is given by:

$$H(X) = \sum_{i=1}^N p_i \log \left(\frac{1}{p_i} \right) = - \sum_{i=1}^N p_i \log(p_i)$$

1.5 Properties of Entropy

1. Entropy is zero if the event is sure (or) impossible.

$$H(X) = 0, \text{ if and only if } p_i = 1, p_k = 0, \text{ for all } k \neq i.$$

2. For a random source X that generates N discrete outputs, the entropy $H(X)$ is bounded by:

$$0 \leq H(X) \leq \log(N)$$

3. When $p_i = (1/N)$ for all N symbol, then the symbols are equally likely. For such case Entropy is given as $H = \log(N)$

Q2: In a binary system if '0' occur with probability 1/4 and '1' occur with probability 3/4. Then calculate amount of information conveyed by each bits.

Ans. Binary symbols: 0 and 1

$$P(0) = \frac{1}{4}, P(1) = \frac{3}{4}$$

The amount of information conveyed by a symbol x is:

$$I(x) = \log_2 \left(\frac{1}{P(x)} \right)$$

Information conveyed by symbol '0'

$$I(0) = \log_2 \left(\frac{1}{1/4} \right)$$

$$I(0) = \log_2(4)$$

$$I(0) = 2 \text{ bits}$$

Information conveyed by symbol '1'

$$I(1) = \log_2 \left(\frac{1}{3/4} \right)$$

$$I(1) = \log_2 \left(\frac{4}{3} \right)$$

$$I(1) = \log_2(1.3333) = 0.415 \text{ bits (approx.)}$$

Q3. If there are N equally likely and independent message, then prove that amount of information carried by each message be $I = n$ bits where $N = 2^n$ and n is an integer.

Ans. If there are N equally likely and independent messages, then the probability of each message is:

$$P = (1/N)$$

The amount of information carried by a message is defined as: $I = \log_2(1/P)$

Substitute $P = (1/N)$,

$$I = \log_2(N)$$

Given in the question: $N = 2^n$. Substitute this into the expression for information:

$$I = \log_2(2^n)$$

Using logarithmic law $\log_2(2^n) = n$:

$$I = n \text{ bits}$$

So, if a source has $N = 2^n$ equally likely messages, then: $I = n$ bits per message.

Q4. Prove the following statement: “If receiver knows the message being transmitted then amount of information carried is zero”

Ans.

The amount of information conveyed by an event (or message) with probability P is defined as:

$$I = \log_2 \left(\frac{1}{P} \right)$$

This formula means: If a message is unlikely, it carries more information. If a message is certain, it carries no new information, because it was already known.

Case: Receiver already knows the message

If the receiver *knows* which message will be transmitted, the probability of that message is: $P = 1$.

Substitute into the information formula:

$$I = \log\left(\frac{1}{1}\right)$$

$$I = 0 \text{ bits}$$

Q5. If I_1 is the information carried by message m_1 and I_2 is the information carried by message m_2 then prove that the amount of information carried composite by m_1 and m_2 is $I_{12} = I_1 + I_2$

Ans.

Here, I_{12} , is the information carried by the combined (composite) message (m_1, m_2) .

For two independent messages, the joint probability is:

$$P(m_1, m_2) = P(m_1) \cdot P(m_2)$$

The information in the composite message is:

$$I_{12} = \log_2\left(\frac{1}{P(m_1, m_2)}\right) = \log_2\left(\frac{1}{P(m_1) \cdot P(m_2)}\right)$$

Using logarithm property:

$$\log_2\left(\frac{1}{ab}\right) = \log_2\left(\frac{1}{a}\right) + \log_2\left(\frac{1}{b}\right)$$

$$\text{So: } I_{12} = \log_2\left(\frac{1}{P(m_1)}\right) + \log_2\left(\frac{1}{P(m_2)}\right)$$

Recognize the terms: $I_{12} = I_1 + I_2$

Thus, the information contained in two independent messages is simply the sum of the individual information contents.

Q6. A source emits one of 4 possible symbol X_0 to X_3 during each signalling interval. The symbol occurs with probability as given in table.

Symbol (X_i)	Probability (P_i)
X_0	0.4
X_1	0.3
X_2	0.2
X_3	0.1

Find the amount of information gained by observing the source emitting each of these symbol and also find the entropy of source.

Ans. Amount of Information for Each Symbol

The self-information of a symbol X_i is: $I(X_i) = \log_2 \left(\frac{1}{P_i} \right)$

For X_0 ($P = 0.4$), $I(X_0) = \log_2 \left(\frac{1}{0.4} \right) = \log_2 (2.5) = 1.322 \text{ bits}$

For X_1 ($P = 0.3$), $I(X_1) = \log_2 \left(\frac{1}{0.3} \right) = \log_2 (3.33) = 1.737 \text{ bits}$

For X_2 ($P = 0.2$), $I(X_2) = \log_2 \left(\frac{1}{0.2} \right) = \log_2 (5) = 2.322 \text{ bits}$

For X_3 ($P = 0.1$), $I(X_3) = \log_2 \left(\frac{1}{0.1} \right) = \log_2 (10) = 3.322 \text{ bits}$

Final Results: Information of Each Symbol

Symbol	Probability	Information $I = \log_2(1/P)$
X_0	0.4	1.322 bits
X_1	0.3	1.737 bits
X_2	0.2	2.322 bits
X_3	0.1	3.322 bits

Entropy of the Source

Entropy is:

$$H = \sum_{i=0}^3 P_i \log_2 \left(\frac{1}{P_i} \right)$$

Substitute values:

$$H = 0.4(1.322) + 0.3(1.737) + 0.2(2.322) + 0.1(3.322)$$

$$H = 0.5288 + 0.5211 + 0.4644 + 0.3322$$

$$H = 1.8465 \text{ bits/symbol}$$

1.6 Information Rate (R)

It is a rate at which information is generated.

$$R = r \cdot H$$

where r = rate at which message is generated, H = Entropy or average information

$R = r$ (message/second) $\times H$ (information bits/message)

Unit of information rate = Information bits/second

Here, r is represented in average number of bits of information per second.

Q7: For a discrete memoryless source with three symbols, the probabilities are $P_1 = \alpha$ and $P_2 = P_3$. Determine the entropy of the source and plot its variation as α changes.

Ans. Consider three symbols m_1, m_2, m_3 with probabilities:

$$P(m_1) = \alpha$$

$$P(m_2) = p$$

$$P(m_3) = p$$

Since total probability = 1:

$$\alpha + p + p = 1$$

$$2p = 1 - \alpha$$

$$p = (1 - \alpha)/2$$

Entropy:

$$H = \alpha \log\left(\frac{1}{\alpha}\right) + p \log\left(\frac{1}{p}\right) + p \log\left(\frac{1}{p}\right)$$

Substitute $p = (1 - \alpha)/2$,

$$H = \alpha \log\left(\frac{1}{\alpha}\right) + 2\left(\frac{1 - \alpha}{2}\right) \log\left(\frac{1}{p}\right)$$

$$H = \alpha \log\left(\frac{1}{\alpha}\right) + (1 - \alpha) \log\left(\frac{2}{1 - \alpha}\right)$$

Final entropy expression:

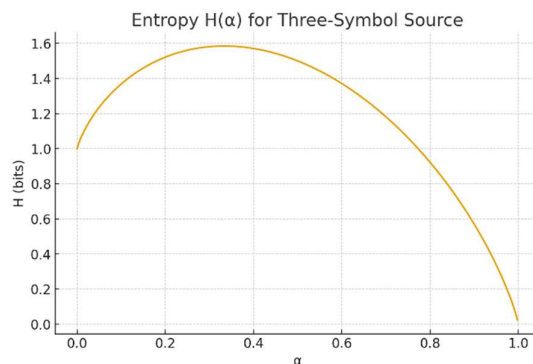
$$H(\alpha) = \alpha \log\left(\frac{1}{\alpha}\right) + (1 - \alpha) \log\left(\frac{2}{1 - \alpha}\right)$$

Variation Table

α	0	0.2	0.3	0.4	0.5	0.6	0.7	0.8	1
H	1	1.5	—	—	1.5	—	—	—	0

Key Property: Entropy is maximum when probabilities are most uniform

For 3 symbols: max occurs near $\alpha = (1/3), p = (1 - \alpha)/2 = (1/3)$



Q8: Find the entropy of a source (in nats per symbol and Hartleys per symbol) that emits one of four symbols A, B, C, and D independently, with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$ respectively.

Ans.

Entropy in NATS per symbol is:

$$H_e = \sum p_i \ln \left(\frac{1}{p_i} \right)$$

Compute each term:

For A:

$$p = \frac{1}{2}, I = \ln 2 = 0.6931$$
$$pI = \frac{1}{2}(0.6931) = 0.3466$$

For B:

$$p = \frac{1}{4}, I = \ln 4 = 1.3863$$
$$pI = \frac{1}{4}(1.3863) = 0.3466$$

For C:

$$p = \frac{1}{8}, I = \ln 8 = 2.0794$$
$$pI = \frac{1}{8}(2.0794) = 0.2599$$

For D: Same as C:

$$pI = 0.2599$$

Total Entropy in NATS

$$H_e = 0.3466 + 0.3466 + 0.2599 + 0.2599$$
$$H_e = 1.213 \text{ nats/symbol}$$

Entropy in Hartleys (base 10 logs) is:

$$H_{10} = \sum p_i \log_{10} \left(\frac{1}{p_i} \right)$$

We use:

$$\log_{10} 2 = 0.3010$$
$$\log_{10} 4 = \log_{10}(2^2) = 0.6021$$
$$\log_{10} 8 = \log_{10}(2^3) = 0.9031$$

For A:

$$pI = \frac{1}{2}(0.3010) = 0.1505$$

For B:

$$pI = \frac{1}{4}(0.6021) = 0.1505$$

For C:

$$pI = \frac{1}{8}(0.9031) = 0.1129$$

For D: same as C

$$pI = 0.1129$$

Total Entropy in Hartleys

$$H_{10} = 0.1505 + 0.1505 + 0.1129 + 0.1129$$

$$H_{10} = 0.527 \text{ Hartleys/symbol}$$

Final Answers

$$H_e = 1.213 \text{ nats/symbol}$$

$$H_{10} = 0.527 \text{ Hartleys/symbol}$$

Q9. A source transmits two independent messages with probabilities P and $1 - P$. Show that the entropy of this binary source is maximum when both messages are equally likely, i.e., when $P = \frac{1}{2}$. Also, plot the entropy of the source as a function of probability P for $0 < P < 1$.

Ans.

1. Entropy of a Binary Source

For two messages m_1 and m_2 with probabilities P and $1 - P$, the entropy is:

$$H(P) = -P \log_2 P - (1 - P) \log_2 (1 - P)$$

This function is symmetric around $P = 0.5$.

2. Condition for Maximum Entropy

To find where entropy is maximum, take the derivative:

$$\frac{dH}{dP} = -\log_2 P - \frac{1}{\ln 2} + \log_2 (1 - P) + \frac{1}{\ln 2}$$

Simplify:

$$\frac{dH}{dP} = \log_2 (1 - P) - \log_2 P$$

Set derivative to zero:

$$\log_2 (1 - P) = \log_2 P$$

$$1 - P = P$$

$$P = \frac{1}{2}$$

Thus, the entropy is maximum when both messages are equally likely.

3. Maximum Entropy Value

$$H_{\max} = -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right)$$

$$H_{\max} = 1 \text{ bit}$$

So the maximum possible entropy of a binary source is 1 bit per symbol.

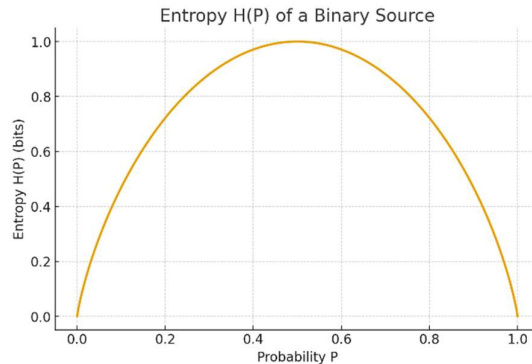
4. Plot of Entropy vs Probability

General behaviour:

At $P = 0 \rightarrow H = 0$

At $P = 1 \rightarrow H = 0$

At $P = 0.5 \rightarrow H = 1$ bit (maximum)



Q10. A telegraph source generates two symbols: dot and dash.

The duration of a dot is 0.2 seconds,

The duration of a dash is three times longer,

The probability of a dot is twice the probability of a dash,

The time gap between consecutive symbols is 0.2 seconds,

Assume a total of 1200 symbols are transmitted.

Determine the information rate (bits per second) of this telegraph source.

Ans.

Step 1: Find symbol probabilities

Let probability of dash = p

Then probability of dot = $2p$

$$2p + p = 1$$

$$3p = 1$$

$$p = \frac{1}{3}$$

Thus:

$$P(\text{dot}) = \frac{2}{3}, P(\text{dash}) = \frac{1}{3}$$

Step 2: Compute entropy per symbol

Binary entropy:

$$H = -\left(\frac{2}{3}\right) \log_2 \left(\frac{2}{3}\right) - \left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right)$$

Compute numerically:

$$H = -\frac{2}{3}(-0.585) - \frac{1}{3}(-1.585)$$

$$H = 0.39 + 0.528 = 0.918 \text{ bits/symbol}$$

Step 3: Find average symbol duration

Dot duration:

$$T_d = 0.2 \text{ sec}$$

Dash duration:

$$T_{dash} = 3T_d = 0.6 \text{ sec}$$

Given inter-symbol gap:

$$T_g = 0.2 \text{ sec}$$

Average symbol duration:

$$\bar{T} = P(dot)(T_d + T_g) + P(dash)(T_{dash} + T_g)$$

$$\bar{T} = \frac{2}{3}(0.2 + 0.2) + \frac{1}{3}(0.6 + 0.2)$$

$$\bar{T} = \frac{2}{3}(0.4) + \frac{1}{3}(0.8)$$

$$\bar{T} = 0.266 + 0.266$$

$$\bar{T} = 0.533 \text{ sec per symbol}$$

Step 4: Symbol rate

$$R_s = \frac{1}{\bar{T}} = \frac{1}{0.533}$$

$$R_s = 1.88 \text{ symbols/sec}$$

Step 5: Information rate

$$R = H \times R_s = 0.918 \times 1.88$$

$$R = 1.726 \text{ bits/sec}$$

Q. 11. A discrete information source emits four symbols with probabilities $P_1 = 0.1, P_2 = 0.2, P_3 = 0.3, P_4 = 0.4$.

(a) Calculate the entropy of the source.

(b) Determine what percentage of the maximum possible entropy is being produced by this source.

Ans.

(a) Entropy of the Source

Entropy is defined as:

$$H = - \sum_{i=1}^4 P_i \log_2 P_i$$

Compute each term:

1. For $P_1 = 0.1$

$$-0.1 \log_2(0.1) = -0.1(-3.3219) = 0.3322$$

2. For $P_2 = 0.2$

$$-0.2 \log_2(0.2) = -0.2(-2.3219) = 0.4644$$

3. For $P_3 = 0.3$

$$-0.3 \log_2(0.3) = -0.3(-1.7369) = 0.5211$$

4. For $P_4 = 0.4$

$$-0.4 \log_2(0.4) = -0.4(-1.3219) = 0.5288$$

Add all terms:

$$H = 0.3322 + 0.4644 + 0.5211 + 0.5288$$

$$H = 1.8465 \text{ bits/symbol}$$

(b) Percentage of Maximum Possible Information

For a source with 4 symbols:

$$H_{\max} = \log_2 4 = 2 \text{ bits/symbol}$$

Percentage output of the source:

$$\begin{aligned} \text{Percentage} &= \frac{H}{H_{\max}} \times 100 \\ &= \frac{1.8465}{2} \times 100 \\ &= 92.325\% \end{aligned}$$

Q12. An image consists of 512×512 pixels, and each pixel can assume one of 8 different intensity levels. Determine the maximum number of bits required to represent the entire image.

Ans.

Step 1: Bits needed per pixel

Each pixel can take 8 levels.

$$8 = 2^3$$

Thus, each pixel requires 3 bits per pixel.

Step 2: Total number of pixels: $512 \times 512 = 262,144$ pixels

Step 3: Total bits required = $262,144 \times 3 = 786,432$ bits.

1.7 n^{th} order extension of the source

- Consider two symbols S_1 and S_2 with probabilities P_1 and P_2 , respectively.
- The second-order extension of this source consists of all possible pairs of symbols.
- Since there are 2 symbols, the number of second-order combinations is: $2^2 = 4$.
- Possible symbol pairs and their probabilities

$$S_1 S_1 : P_1 \cdot P_1 = P_1^2$$

$$S_1 S_2 : P_1 \cdot P_2$$

$$S_2 S_1 : P_2 \cdot P_1$$

$$S_2 S_2 : P_2 \cdot P_2 = P_2^2$$

Q.13. Prove n^{th} order extension of entropy: $H(S^n) = n \cdot H(S)$.

Ans. Consider a discrete memoryless source (DMS) with alphabet: $S = \{s_1, s_2, \dots, s_m\}$ and corresponding symbol probabilities: $P(s_i)$, $i = 1, 2, \dots, m$. The entropy of the zero-order (original) source is:

$$H(S) = - \sum_{i=1}^m P(s_i) \log_2 P(s_i)$$

n-th Order Extension of the Source: An n-th order extension combines n successive symbols to form a new “super-symbol.” Since the source is memoryless, occurrences of symbols are independent.

So, for a sequence, $(s_{i_1}, s_{i_2}, \dots, s_{i_n})$, The joint probability of that sequence is:

$$P(s_{i_1}, s_{i_2}, \dots, s_{i_n}) = P(s_{i_1}) P(s_{i_2}) \dots P(s_{i_n})$$

Entropy of n-th Order Source: The entropy of an extended symbol (of length n) is:

$$H(S) = - \sum_{\text{all sequences}} P(s_{i_1}, s_{i_2}, \dots, s_{i_n}) \log_2 P(s_{i_1}, s_{i_2}, \dots, s_{i_n})$$

Substitute joint probability:

$$P(s_{i_1}, s_{i_2}, \dots, s_{i_n}) = \prod_{k=1}^n P(s_{i_k})$$

$$H(S^n) = - \sum_{\text{all sequences}} \left(\prod_{k=1}^n P(s_{i_k}) \right) \log_2 \left(\prod_{k=1}^n P(s_{i_k}) \right)$$

Apply Log of a Product

$$\log_2 \left(\prod_{k=1}^n P(s_{i_k}) \right) = \sum_{k=1}^n \log_2 P(s_{i_k})$$

Thus:

$$H(S^n) = - \sum_{\text{all sequences}} \left(\prod_{k=1}^n P(s_{i_k}) \right) \left(\sum_{k=1}^n \log_2 P(s_{i_k}) \right)$$

Reverse sum order:

$$H(S^n) = - \sum_{k=1}^n \sum_{\text{all sequences}} \left(\prod_{j=1}^n P(s_{i_j}) \right) \log_2 P(s_{i_k})$$

Notice the Key Fact: Each inner sum corresponds to the entropy of the original source, because for each position:

$$\sum_{\text{symbols}} P(s_{i_k}) \log_2 P(s_{i_k})$$

occurs independently, and across n positions. Thus:

$$H(S^n) = n \left(- \sum_{i=1}^m P(s_i) \log_2 P(s_i) \right)$$

$$H(S^n) = n \cdot H(S)$$

The entropy of the n -th order extension of a memoryless source equal to n times the entropy of the original source, because symbols are independent and information adds up across each symbol.

Q.14. A discrete memoryless source has an alphabet $S = \{S_0, S_1, S_2\}$, with corresponding symbol probabilities $P(S_0) = 0.7, P(S_1) = 0.15, P(S_2) = 0.15$.

(a) Compute the entropy of the source.

(b) Determine the entropy of the second-order extension of this source.

Ans.

(a) Entropy of the Source

The entropy of a discrete memoryless source is:

$$H = - \sum_i P_i \log_2 P_i$$

Substitute probabilities:

$$H = -[0.7 \log_2(0.7) + 0.15 \log_2(0.15) + 0.15 \log_2(0.15)]$$

$$H = -[0.7(-0.5145) + 0.15(-2.7369) + 0.15(-2.7369)]$$

$$H = -[-0.3601 - 0.4105 - 0.4105]$$

$$H = 1.1811 \text{ bits/symbol}$$

(b) Entropy of the 2nd-Order Extension

For a zero-order (memoryless) source: Symbols are independent.

Entropy of a 2-symbol block is:

$$H^{(2)} = 2H$$

Thus:

$$H^{(2)} = 2 \times 1.1811 = 2.3622 \text{ bits per 2-symbol block}$$